

Efficient and Robust SEM Image Denoising for Wafer Defect Inspection

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Abstract

Noise in scanning electron microscopy (SEM) often obscures details critical for accurate wafer defect inspection. Deep learning-based denoising methods have been widely used to address this problem, but they have two major limitations in SEM image denoising: lack of both efficient and powerful denoising methods, and poor generalization to image structures that are unseen during training. To this end, we propose Relaxed Noise2Noise with Input dropout (ReNIn), which includes components that address the above two issues. Firstly, our Relaxed Noise2Noise (RN2N) framework provides a much better trade-off between denoising performance and training data collection cost; namely, it shows nearly on-par denoising performance with much lower data collection cost than ordinary supervised learning-based methods. Secondly, we propose to apply the *input dropout* to boost generalization ability. It improves the performance of the images that are structurally different from the training images without compromising the performance of the normally structured images, thereby increasing the overall denoising performance. Consequently, our method supports downstream inspection tasks by reducing the failure rate in circle detection, which is a critical preprocessing step for circle-shaped product analysis. Overall, our ReNIn attains efficient training data collection cost with competitive denoising performance and enhances generalization capability across various structures.

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Introduction

The Scanning Electron Microscope (SEM) has served as a standard tool for identifying manufacturing defects in semiconductor wafers (Dey et al., 2021). The raw images captured from the SEM are often degraded by noise, making denoising an essential step for fine-grained defect inspection. Traditionally, SEM image denoising is achieved by capturing multiple frames of the same image and averaging them; a method that is highly expensive, time-consuming, and potentially damaging the image samples (Giannatou et al., 2019). To address these limitations, several deep learning-based SEM image denoising techniques have been developed, which only require forward passes of neural networks during denoising.

Following the promising results of convolutional neural network-based denoisers (Burger et al., 2012; Mao et al., 2016; Zhang et al., 2017; Tai et al., 2017; Lefkimmiatis, 2018) on natural noisy images from standard cameras, initial studies on SEM image denoising (Chaudhary et al., 2019; Ede and Beanland, 2019; Giannatou et al., 2019; Nagano et al., 2021) adopt the supervised learning framework, which uti-

lizes clean-noisy image pairs for training. Despite promising results, obtaining sufficient clean-noisy SEM image pairs for training is highly expensive. To lift the dependency on obtaining clean SEM images, recent work (Yu et al., 2020; Sato et al., 2022) have adopted the Noise2Noise (N2N) (Lehtinen et al., 2018) training scheme in which two independent realizations of noisy images for the same source are utilized in training. Despite the cheap data acquisition cost, N2N shows inferior denoising performance compared to its supervised counterpart in SEM image denoising, as illustrated in Figure 1. Note that the supervised denoiser (Sup) and N2N are positioned at extremes where denoising performance and training data acquisition cost are either high or low. Specifically, when a raw noisy image (denoted by F01) is used as the input of the model, the supervised denoiser uses a highly clean target image, which can be typically obtained by averaging 64 frames of raw noisy images (denoted by F64), while N2N uses a single raw noisy image (F01) as a target.

As the supervised denoiser and N2N represent two extremes in data collection cost and denoising performance, we believe there is a middle ground between them. To find a

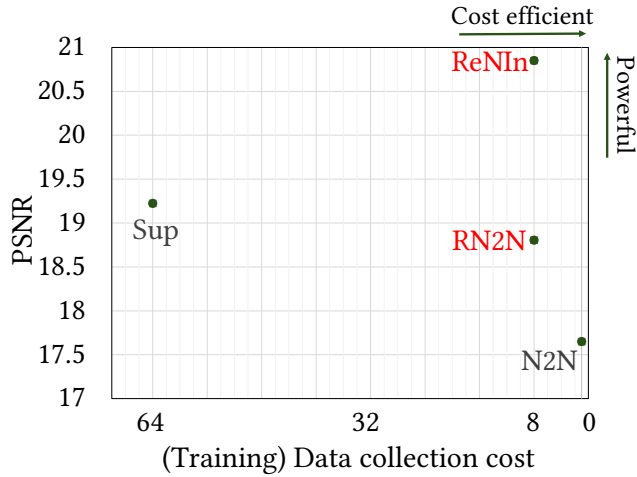


Fig. 1. Training data collection cost vs. Denoising performance (PSNR). We compare the models on our semiconductor wafer dataset (described in [SEM images dataset](#) section) using PSNR as the evaluation metric. Each number on the x-axis (training data collection cost) represents the number of raw noisy images (F#) required to obtain a single target image. For example, for a supervised denoiser (denoted by Sup) 64 frames of the raw noisy images are obtained and then averaged to get a single image (F64), which is regarded as a clean image. In the plot, a model is considered efficient and powerful as the point moves toward the upper-right direction. Here, raw noisy images (F01) are used as input for all models. **RN2N** and **ReNIn** denote our proposed methods.

better trade-off, we devise the Relaxed Noise2Noise (RN2N), which relaxes the noise distribution constraint on the target image. For example, instead of using a highly clean image (F64) or highly noisy image (F01) as a target, we use a moderately clean image, such as an 8-frame averaged image (denoted by F08), as the target. The preview of the impact of our RN2N is shown in [Figure 1](#) — it is evident that RN2N significantly reduces data collection cost (8× cheaper) while achieving competitive performance with its supervised counterpart (with a gap smaller than 0.5 dB).

Another challenge for SEM image denoising is poor generalization ability to image structures that are unseen during training. Namely, N2N shows decent denoising performance when test images are structurally similar to the training images. However, when the image structures of test images are significantly different from those of training images, N2N fails to successfully denoise them. For example, as depicted in [Figure 2](#), the N2N model trained on regular circular images shows poor generalization ability when applied to test images with irregular and bumpy circles — note it particularly fails to preserve the edges of circles. Given the nature of the semiconductor manufacturing process where SEM images with irregular structures are frequently captured, handling *structurally different* test images is crucial for semiconductor SEM image denoising. To tackle this, we apply the

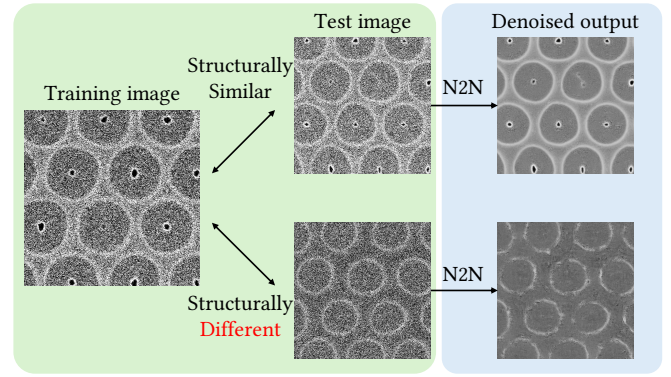


Fig. 2. Generalization issue of N2N. The green box shows the examples of the training SEM image along with two types of test images. In the green box, the upper right image represents a test image that is structurally similar to the training image (regular circles), while the lower right image denotes a structurally different test image (irregular and bumpy circles). The blue box represents the denoised images obtained from N2N. N2N fails to denoise the structurally different test image effectively, particularly in preserving edges.

input dropout mechanism which randomly drops pixels on the input image during training. Our experimental results demonstrate that it considerably enhances the conventional denoising metrics (PSNR/SSIM) but also in detecting circles, which is a crucial step for subsequent semiconductor defect inspection.

In summary, our final model, Relaxed Noise2Noise with Input Dropout (ReNIn), successfully addresses two major issues in semiconductor SEM image denoising: lack of efficient training consideration and poor generalization ability. In [Figure 1](#), we indeed observe that ReNIn largely enhances the denoising performance over RN2N, even surpassing that of supervised denoiser, with a much lower data collection cost.

Related Works

[Deep learning-based denoisers without clean image]

As mentioned above, to remove the requirements of clean images for supervised training, several attempts have been proposed denoisers which can solely be trained with noisy images in the natural image domain. The first approach adopts the Noise2Noise (N2N) training scheme ([Lehtinen et al., 2018](#)) in which two independent realizations of noisy images are utilized in training. This approach demonstrates competitive denoising performance with supervised counterparts in the natural image domain. However, the necessity of two noisy realizations of the exact same source can still be impractical in real-world settings. Therefore, another approach adopts a self-supervised learning-based denoising framework ([Batson and Royer, 2019](#); [Laine et al., 2019](#); [Krull](#)

et al., 2019; Moran et al., 2020; Xu et al., 2020; Wu et al., 2020; Byun and Moon, 2020; Quan et al., 2020; Huang et al., 2021; Byun et al., 2021), which utilizes only *single* noisy images itself for training. To enable self-supervision, most works have adopted the blind spot network (BSN), which prevents the model from simply learning identity mapping by masking the central pixel in the receptive field of the noisy image. Recently, FBI-denoiser (Byun et al., 2021) achieved superior performance on real-world natural noisy image benchmark datasets and Fluorescence Microscopy Denoising dataset (Zhang et al., 2019). It proposed an efficient BSN architecture (named FBI-Net) for pixel-wise affine denoiser and a separate network that estimates Poisson-Gaussian noise parameters. A similar work to our Relaxed Noise2Noise is Moran et al. (2020). It uses two different noisy images with different noise levels for training. Their approach is to generate additional noisy images by adding known noise to single noisy images. It is similar to ours in that it uses two different noisy images, each with noise from different distributions. It assumes that the noise comes from a known noise model. However, the noise modeling for our SEM image is not straightforward since our images are obtained directly from the SEM. In addition, it is further complicated by the possible presence of spatially correlated noise, which will be addressed further in Discussion section. It makes it impractical to apply their approach to our SEM images. Our method, on the other hand, requires additional noisy images without any constraints rather than hard-to-define precise noise information.

The presence of correlated noise also demonstrates why the denoising results of the above second approach (the self-supervised denoisers) are significantly degraded in our experiments. Thus, rather than self-supervised denoisers, we select the N2N-based training approach (Lehtinen et al., 2018) as the main baseline in our experiments. In this work, between the N2N approach and its supervised counterpart, we aim to find a better trade-off in terms of performance and data collection cost.

[Overfitting issues in low-level vision] Several works (Quan et al., 2020; Kong et al., 2022; Liu et al., 2023; Chen et al., 2023; Ma et al., 2024; Gu et al., 2024) have been proposed to mitigate overfitting issues in low-level vision tasks. For example, the dropout technique (Srivastava et al., 2014), which randomly deactivates neurons during training, has been applied to handle overfitting in *single* image denoising (Quan et al., 2020) and super-resolution (Kong et al., 2022). Similarly, recent studies (Chen et al., 2023; Ma et al., 2024) have adopted masked autoencoder (MAE), a network architecture designed to reconstruct missing or corrupted input data, to mitigate overfitting of the model to the noise distribution during training. Our proposed input dropout mechanism is closely related to both the dropout technique and MAE, but, to the best of our knowledge, it is the first application of an input masking scheme specifically focused on

enhancing the generalization ability for image structures in SEM image denoising.

Preliminaries and Notations

Supervised Learning

In supervised learning, the D pairs of noisy-clean images $\{\mathbf{y}^i, \mathbf{x}^i\}_{i=0}^D$ are used, where the i -th noisy image is referred to as $\mathbf{y}^i$ and the corresponding clean image as $\mathbf{x}^i$. Its training objective with a learnable parameter θ for the model is given as

$$\arg \min_{\theta} \sum_i L(f_{\theta}(\mathbf{y}^i), \mathbf{x}^i) \quad (1)$$

in which f_{θ} is the neural network-based denoiser, and L denotes the loss function, which typically adopts the L_2 loss, also known as the MSE (Mean Squared Error) loss.

Noise2Noise

Instead of using the noisy-clean image pairs for training, two noisy image pairs $\{\mathbf{y}^i, (\mathbf{y}')^i\}_{i=0}^D$ are utilized, where $(\mathbf{y}')^i$ is another noisy realization sampled from the same underlying clean image $\mathbf{x}^i$ of $\mathbf{y}^i$. The key difference between $\mathbf{y}^i$ and $(\mathbf{y}')^i$ lies in their noise realizations, which are assumed to be independent and identically distributed (i.i.d.) with zero mean. Under this assumption, training the neural network with these two noisy images enables the model to learn the statistical mean of the noisy images. The training objective for training the denoiser f_{θ} with Noise2Noise is as follows:

$$\arg \min_{\theta} \sum_i L(f_{\theta}(\mathbf{y}^i), (\mathbf{y}')^i) \quad (2)$$

While its approach relies on the assumptions of zero mean and i.i.d. noise, experimental results in (Lehtinen et al., 2018, Section 3.2 Other Synthetic Noises) demonstrate that Noise2Noise can effectively denoise even when the noise does not strictly adhere to these assumptions, such as in the case of text overlay. Additionally, it demonstrates decent denoising performance—comparable to supervised learning-based approaches in certain settings—while significantly reducing data acquisition costs by eliminating the need for clean images during training. However, in extremely noisy cases with complex noise characteristics, such as in our SEM dataset, it is not clear whether Noise2Noise would still achieve competitive denoising performance compared to supervised methods.

Materials and Methods

In this section, we will describe our main contributions: we first present the details of the semiconductor SEM image dataset we used, then delineate the intuition and details of Relaxed Noise2Noise with Input dropout (ReNIn).

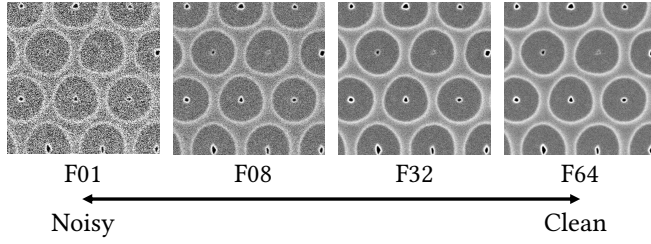


Fig. 3. Examples of our SEM images with different noise levels. The ground truth clean images (F64) are obtained by averaging 64 raw images (F01). As more frames are used for averaging (moving to the right in the figure), the image becomes less noisy.

SEM images dataset

Our SEM images consist of noisy images from eight distinct scenes, each with a resolution of 3072x2048 pixels. The term “scene” refers to the different view of NAND flash semiconductor products, which are core components of Solid State Disk (SSD). These images are sourced from the manufacturing systems of Samsung Electronics, Inc. Within each scene, noisy images at four different noise levels (F01, F08, F32, and F64, as shown in Figure 3) are obtained by adjusting the number of frames averaged during acquisition. Namely, the number following “F” indicates the number of frames used. For example, F01 represents a single-frame (raw) image captured directly from the SEM, while F08 corresponds to images with reduced noise, achieved by averaging 8 frames. F32 images are primarily used to represent the upper bound performance of the median filter, as applying the median filter to F01 images often yields undesirable results. Despite the higher data collection cost, the median filter on F32 is preferred in practical manufacturing due to its stability and high performance. F64 images are used as ground-truth clean images in supervised learning and evaluation. Depending on the training scheme, the different F# target images are selectively used for training. For instance, for the target images, the supervised method uses F64, the self-supervised method uses the same F01 itself, and N2N uses a different F01 with a distinct noise realization. For all methods, F01 images are selected as input due to their high efficiency, eliminating the need to capture multiple frames in the manufacturing systems.

The scenes can be categorized into two structural types: six scenes with normal structures and two with *structurally different* structures. As shown in the bottom row of Figure 2, the *structurally different* F# images, abbreviated as SDF#, have oval or dented circles that differ from the precisely circular F01 images observed in the normally structured training image. Additionally, the small circle pattern within the larger circle is often barely visible, in contrast to the normal structure. A significant challenge with these *structurally different* images is that the N2N denoiser, trained exclusively

on normal F01 images, struggles to process them effectively, as highlighted in the blue box of Figure 2. Among the eight scenes, five normally structured scenes are selected to form the training dataset, and three remaining scenes are selected as the test dataset to assess the performance and generalization ability of the denoising model: one from the normal structure and two from the *structurally different* scenes. For simplicity, these test scenes are referred to as 1st F01, 1st SDF01, and 2nd SDF01 in the experimental results. The key difference between the two SDF01 images is that the 2nd SDF01 image has a much smaller dented circle than the 1st SDF01. Since the model is trained only on normal structure images, successful denoising of the three evaluation scene images indicates that the model effectively handles both types of images, demonstrating good generalization ability.

Moreover, to resolve a spatial misalignment problem exists due to the inevitable motion shaking of the SEM and specimen during image acquisition, we devise a straightforward PSNR metric-based image registration algorithm, which is described in the Appendix (PSNR matching section).

Relaxed Noise2Noise with Input dropout (ReNIn)

Now, we explain our approach, Relaxed Noise2Noise with Input dropout (ReNIn), specifically designed to enhance generalization while maintaining efficient training data collection cost for SEM image denoising.

[Relaxed Noise2Noise (RN2N)] As described in Preliminaries and Notations section, Noise2Noise requires training image pairs to have noise realizations that are independent and identically distributed, meaning they must have the same noise statistics. However, as shown in Figure 1, we believe that using moderately clean images can provide a better trade-off between data collection cost and denoising performance. Based on this idea, we propose to relax the constraint of Noise2Noise by allowing training with noisy images from the same noise distribution but with *different* noise statistics (more specifically, different noise variance). For example, when the input noisy image is F01, a relatively cleaner target image like F02 or another F# image can be used instead of using a target image with the same noise statistics (y' in Equation 2). In our experiments, we use F08 images as target.

[Input dropout (ID)] In the existence of *structurally different* images, N2N shows degraded performance on them. In addition, the denoised images of N2N and the supervised method have trouble constructing connected edges, as shown in Figure 7. We conjecture that this is mainly due to overfitting to a specific structure of images and extreme noise levels that make regions between edges and others ambiguous. To handle such problems, we propose an input dropout (ID) strategy, which applies dropout directly to the input pixels, rather than to neurons in hidden layers as in

Table 1. PSNR (dB)/SSIM results of ReNIn and application of regular dropout. PSNR and SSIM values are averaged over three test F01 images. “ID” refers to our input dropout strategy, while “CD” refers to the conventional dropout strategy, which drops neurons in hidden layers. D1 to D5 indicate the positions within the FBI-Net architecture where dropout is applied. The F01-F08 model (RN2N) is used as the base model. More details of the dropout layer position and experiment can be found in [Dropout experiment details](#) section.

Model	PSNR	SSIM
RN2N	18.8050	0.2400
RN2N + ID (ReNIn)	20.8514	0.2712
RN2N + CD (D1)	18.6695	0.2329
RN2N + CD (D2)	18.7859	0.2384
RN2N + CD (D3)	18.8553	0.2393
RN2N + CD (D4)	18.7790	0.2362
RN2N + CD (D5)	18.7836	0.2351



Fig. 4. Overfitting issue of N2N. The green and red graphs represent the test and training loss, respectively. The intersection of these two graphs clearly indicates an overfitting issue of the N2N model.

conventional dropout strategies. Namely, the pixels of the image are dropped by multiplying a binary random mask $\mathbf{m}$, generated by i.i.d. Bernoulli distribution with the dropout rate p . With the masked image, the model is trained to perform dual tasks: removing the noise present in unmasked pixels while predicting the values of the masked pixels. As shown in [Table 1](#), conventional dropout (rows 3 to 7) shows no improvement, while our input dropout significantly enhances generalization.

Overall, for each pair of $\{\mathbf{y}^i, (\mathbf{y}')^i\}_{i=0}^D$, the training objective of ReNIn is:

$$\arg \min_{\theta} \sum_i L(f_{\theta}(\mathbf{y}^i \odot \mathbf{m}^i), (\mathbf{y}')^i) \quad (3)$$

in which $\mathbf{m}^i$ is the pixel-wise mask for $\mathbf{y}^i$, $\odot$ is the element-wise multiplication, and $(\mathbf{y}')^i$ is the (relatively clean) target

image. In our experiments, we mainly use F01 images as input ($\mathbf{y}$) and F08 images as target ($\mathbf{y}'$).

After training, the denoiser takes test images without applying input dropout and returns its expected clean image. The estimated clean image $\hat{\mathbf{x}}^t$ is denoted as :

$$\hat{\mathbf{x}}^t = f_{\theta}(\mathbf{y}^t) \quad (4)$$

where $\mathbf{y}^t$ is the test image.

Experimental Results

Experimental settings

Following previous studies ([Brooks et al., 2019](#); [Cheng et al., 2021](#); [Yang et al., 2024](#)), training images are randomly cropped to 256x256 pixels, resulting in a total of 21,600 pairs ($D = 21,600$) of patch. Random horizontal and vertical flips are also applied. For evaluation, the original size images (3072×2048) are used.

Baseline We have conducted experiments and comparisons with several denoising methods, including the Median filter, BM3D ([Dabov et al., 2007](#)), Noise2Void (N2V) ([Krull et al., 2019](#)), FBI-denoiser ([Byun et al., 2021](#)), Noise2Noise (N2N) ([Lehtinen et al., 2018](#)), and the supervised learning method. These methods are categorized into conventional denoising methods (Conventional) and deep learning-based methods.

For conventional denoising, we evaluate three methods: the median filter (F01), the median filter (F32), and BM3D. The median filter is a training-free approach widely used in real-world manufacturing for its stable and edge-preserving results, especially for the already *relatively clean* images like F32. The input image used for each median filter method is indicated in parentheses. As expected, the median filter (F01) produces suboptimal results, while the median filter (F32) serves as an upper bound for denoising performance, assuming *expensive* F32 images are available during denoising. As another baseline, we use BM3D ([Dabov et al., 2007](#)), recognized as the state-of-the-art among classical denoising methods. Since BM3D requires specifying the noise standard deviation (σ), we report results with $\sigma = 140$, which achieves the best PSNR.

We evaluate several deep learning-based methods: Noise2Void ([Krull et al., 2019](#)) and FBI-denoiser ([Byun et al., 2021](#)), Noise2Noise ([Lehtinen et al., 2018](#)), and a supervised learning method. To ensure fair comparisons among them, we used the same BSN architecture, FBI-Net, proposed in ([Byun et al., 2021](#)) for all methods. Noise2Void and FBI-Denoiser are the self-supervised learning baselines, which are trained on single noisy input images using BSN — the difference is that the FBI-Denoiser estimates affine mapping coefficients for the input images to generate the denoised output, while Noise2Void directly estimate the denoised output. On the other hand, Noise2Noise and supervised learning use target images that are separate from the input for

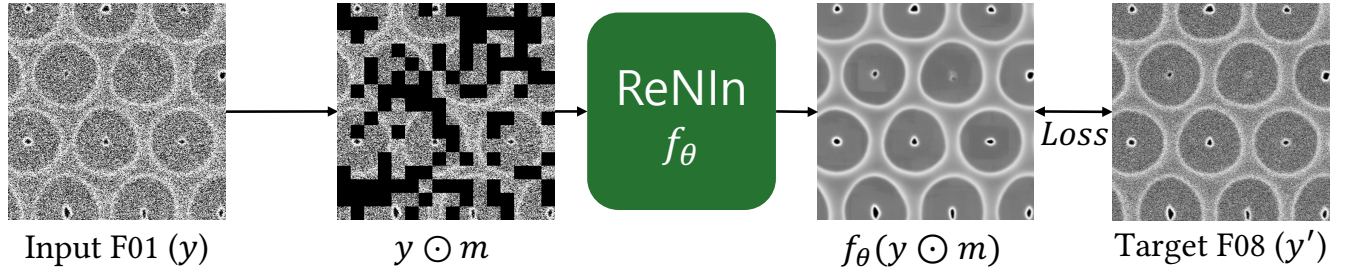


Fig. 5. Overall training procedure of ReNIn. Here, $\odot$ denotes element-wise multiplication. Note that the masking strategy is only applied in the training phase, not during the inference phase.

training, and the only difference between them in our setting is the noise level of the target images; *i.e.*, Noise2Noise uses a separate noisy F01 image as a target and the supervised learning uses F64 as a “clean” target. Note both of these baselines do not necessarily have to use the BSN, but we fixed to use the BSN architecture so as to directly compare the performance with the self-supervised learning baselines. Moreover, as we show in our [Ablation studies on modeling choices](#), the performance of Noise2Noise turns out to not differ significantly even when we use UNet (Ronneberger et al., 2015), which has a non-BSN architecture.

Evaluation metric

The evaluation is based on two standard image quality metrics, and a failure rate metric in the circle detection task which identifies the edges of circular patterns in each SEM image—a crucial pre-processing step in the manufacturing analysis and inspection.

[PSNR and SSIM] Image quality assessment is conducted using two standard metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), utilizing the F64 images that correspond to the input F01 images as clean reference. As mentioned in [SEM images dataset](#), three scenes are selected for sampling test images: a normally structured single F01 image and two structurally different F01 (SDF01) images.

[Failure rate in circle detection] We utilize the circle detection failure rate—defined as the proportion of undetected circles relative to the total number of circles—as a key evaluation metric. To calculate this metric, circle detection is performed separately for each circle in the image. Specifically, a human-annotated ground-truth image indicating the true circle regions is used to identify the region corresponding to each circle. Based on these identified regions, circle detection is conducted for each circle individually. We do not additionally consider the false negative rate since there is exactly one circle to detect, reducing the task to determining whether the circle is detected within its expected region.

A lower failure rate indicates that denoised images retain clear structural information around edges, vital for dis-

tinguishing between cut and uncut parts in semiconductor manufacturing. Therefore, regardless of the above image quality, we believe it clearly shows the effectiveness of the denoising process which has a crucial role in identifying manufacturing defects in semiconductor wafers. It would contribute to subsequent product inspection and analysis, such as critical dimension or line edge roughness measurement. The failure rate (FR) for the test image y^t is represented as :

$$FR(y^t) = \frac{C_{fail}}{C} \times 100 (\%) \quad (5)$$

where C is the number of total circles to capture in the test images y^t , and C_{fail} is the number of circles which fail to detect. In our experiments, three scenes are used and 108 circles ($C = 108$) for each scene are captured for evaluation.

Main results

[Performance of ReNIn] As shown in [Table 2](#), [Figure 6](#), and [Figure 7](#), ReNIn shows superior denoising results in terms of PSNR, SSIM, and visual quality. In [Table 2](#), it achieves an averaged PSNR of 20.85 dB, showing a significant improvement in denoising performance, particularly with the *structurally different* images. It even surpasses the PSNR of supervised learning which is about 19.22 dB, mainly due to its enhanced performance on SDF01 images. Specifically, ReNIn almost reaches the performance of supervised learning for 1st F01 images within 0.61 dB difference in PSNR at 8× cheaper cost (F64 → F08) for collecting target training data. For SDF01 images, ReNIn achieves 20.39 dB and 19.12 dB, respectively. These values are significantly higher compared to the PSNR results from supervised learning, which are approximately 16.24 dB and 17.77 dB. We attribute this performance enhancement on SDF01 images to the superior generalization ability of ReNIn, supported by input dropout. The effectiveness of ReNIn is further validated by qualitative results for both normal and *structurally different* images, as shown in [Figure 6](#) and [Figure 7](#). On normal structure images ([Figure 6](#)), our method ([Figure 6j](#)) and deep learning-based methods except for self-supervised learning methods

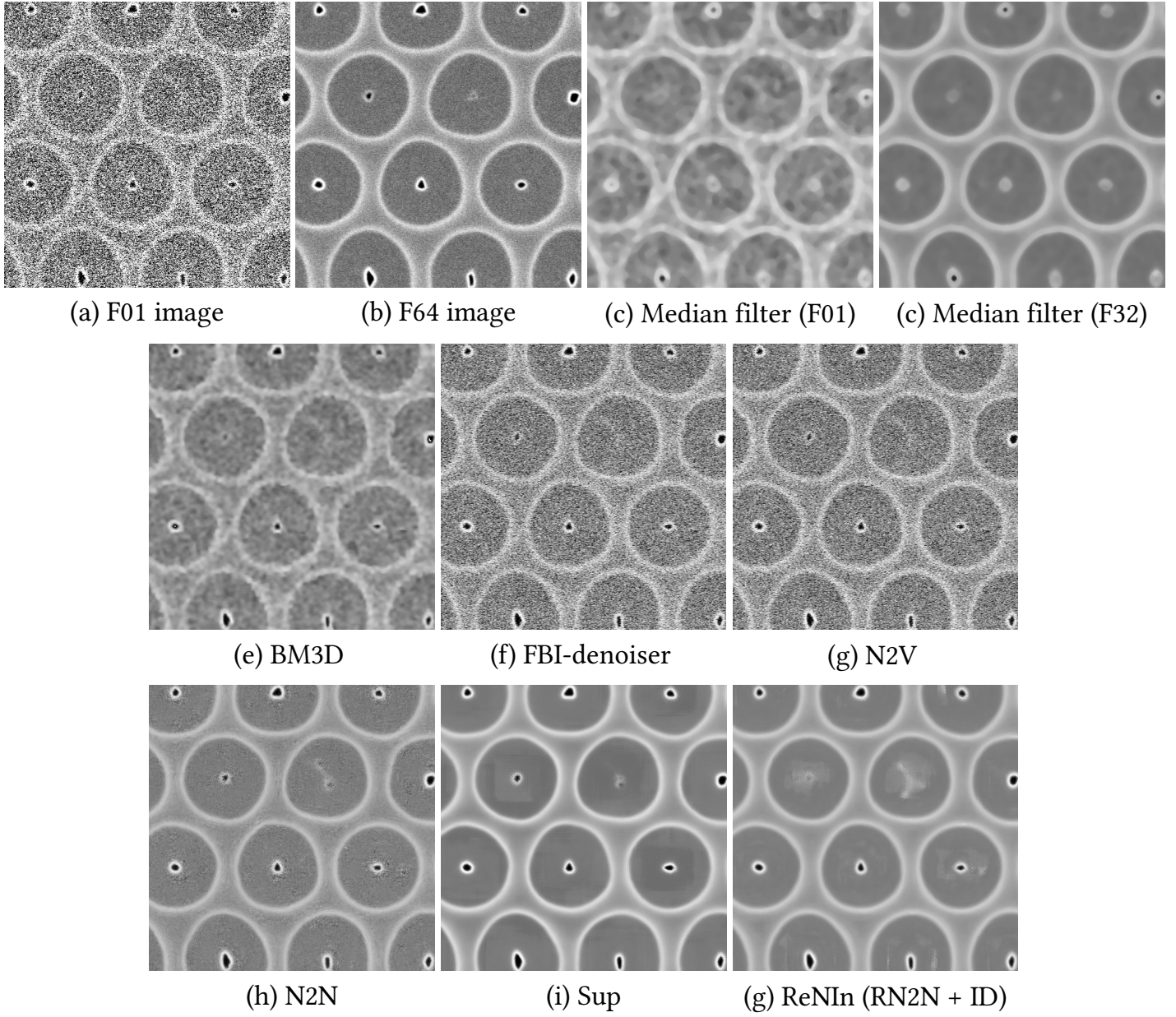


Fig. 6. Qualitative denoising results on a structurally similar test image. Here, the input test image is structurally similar to the training image.

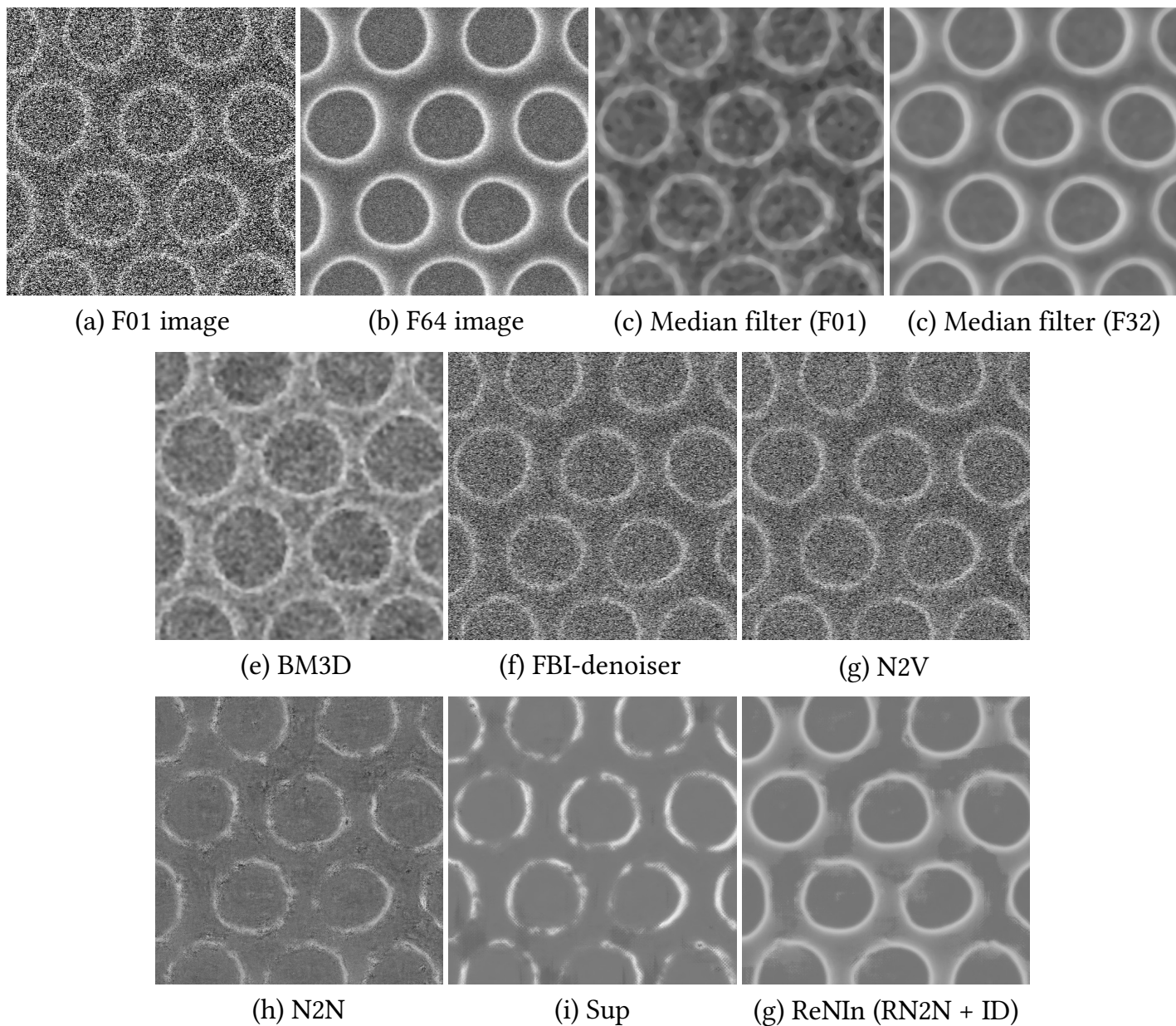


Fig. 7. Qualitative denoising results on a *structurally different* test image (SDF01). Here, the input test image is structurally different from the training image. The results of another SDF01 image are omitted for simplicity since it shows a similar tendency.

Table 2. PSNR(dB)/SSIM results of various methods. Since the evaluation images consist of one F01 and two SDF01 images, they are referred to as 1st F01, 1st SDF01, and 2nd SDF01 for convenience. The “average” column in PSNR/SSIM shows the averaged PSNR/SSIM values of three evaluation images. The best results for each image are marked in **bold** while the second ones are underlined except for the median filter (F32) since it utilized high-cost F32 images as input.

category	model	PSNR / SSIM ($\uparrow$)			
		1st F01	1st SDF01	2nd SDF01	average
Conventional	Median filter (F32)	22.64 / 0.3382	21.70 / 0.2616	21.19 / 0.2409	21.84 / 0.2802
	Median filter (F01)	20.44 / 0.3143	14.09 / 0.2206	17.56 / 0.2130	17.36 / 0.2493
	BM3D (Dabov et al., 2007)	21.34 / 0.2974	15.12 / 0.2008	17.70 / 0.1905	18.05 / 0.2296
Deep learning-based	Noise2Void (Krull et al., 2019)	14.56 / 0.0792	12.20 / 0.0500	13.35 / 0.0540	13.37 / 0.0611
	FBI-denoiser (Byun et al., 2021)	14.71 / 0.0827	12.30 / 0.0505	13.42 / 0.0537	13.48 / 0.0623
	Noise2Noise (Lehtinen et al., 2018)	21.92 / 0.2990	14.40 / 0.1150	16.63 / 0.1403	17.65 / 0.1848
	Supervised learning	23.66 / 0.3668	<u>16.24 / 0.1727</u>	<u>17.77 / 0.1830</u>	<u>19.22 / 0.2408</u>
	ReNIn (Ours)	<u>23.05 / 0.3446</u>	20.39 / 0.2508	19.12 / 0.2181	20.85 / 0.2712

Table 3. Failure rate (FR) results of various methods. Details are the same as Table 2.

category	model	FR($\downarrow$)			
		1st F01	1st SDF01	2nd SDF01	average
Conventional	Median filter (F32)	0.00%	0.00%	0.00%	0.00%
	Median filter (F01)	12.50%	36.57%	<u>52.31%</u>	<u>33.80%</u>
	BM3D (Dabov et al., 2007)	8.33%	<u>34.26%</u>	67.59%	36.73%
Deep learning-based	Noise2Void (Krull et al., 2019)	<u>0.93%</u>	96.30%	87.04%	61.42%
	FBI-denoiser (Byun et al., 2021)	1.85%	96.30%	83.33%	60.49%
	Noise2Noise Lehtinen et al. (2018)	0.00%	100.00%	100.00%	66.67%
	Supervised learning	0.00%	100.00%	100.00%	66.67%
	ReNIn (Ours)	0.00%	0.93%	2.31%	1.08%

(Figure 6h and Figure 6i show good visual results while self-supervised learning methods and other conventional denoising approaches fall short. Especially, the denoised result of the median filter (F32) looks blurry compared to ReNIn although it shows superior performance in PSNR and SSIM. On *structurally different* images (Figure 7), other methods struggle with noise removal, as seen in Figure 7g, or fail to preserve edge information, as observed in Figure 7i. In contrast, our method effectively removes noise while preserving outer edge details, as depicted in Figure 7j, achieving results comparable to the median filter (F32), which uses expensive F32 images. We believe this significant improvement in handling challenging images underscores the strong generalization ability of ReNIn.

Regarding the failure rate (FR) of the circle detection metrics, ReNIn achieves superior circle detection performance compared to other baselines, as shown in Table 3 and Figure 8. These results highlight its strong generalization ability across various image structures. Specifically, ReNIn achieves a significantly lower average failure rate (FR) of 1.08%, whereas Noise2Noise struggles with effective circle detection, resulting in an average FR of 66.67%. In the visualization results presented in Figure 8, we again observe that Noise2Noise fails to perform well with *structurally different* images (Figure 8b), in contrast to its performance with normal images (Figure 8a). Whereas, our ReNIn performs con-

sistently well in circle detection, regardless of whether the images are normal images (Figure 8c) or *structurally different* images (Figure 8d).

[The inferior performance of other baselines] As shown in Table 2, Figure 6 and Figure 7, the baseline methods including BM3D, Noise2Noise, N2V, and FBI-denoiser do not perform well. We conjecture that it is mainly due to the characteristics of our SEM dataset, highlighting the challenges it presents. For BM3D, we believe its limited performance compared to other methods stems from the *source-dependent characteristic* of our SEM image dataset. Namely, BM3D is mainly designed to handle *source-independent* noise, such as Gaussian noise. For Noise2Noise, it does not show comparable performance to the supervised learning method, unlike previous studies (Lehtinen et al., 2018; Zhang et al., 2019). We attribute this to extremely noisy images of our SEM dataset compared to the images used in the previous studies, leading to a performance gap between the N2N and supervised learning method shown in Figure 1. In contrast, our ReNIn shows improved performance thanks to the use of moderately noisy F08 images as target images and improved generalization with suggested input dropout. We also observe that the self-supervised methods (N2V and FBI-Denoiser), which assume pixel-wise independent noise, show inferior denoising performance compared to other baselines. This result suggests the possible presence

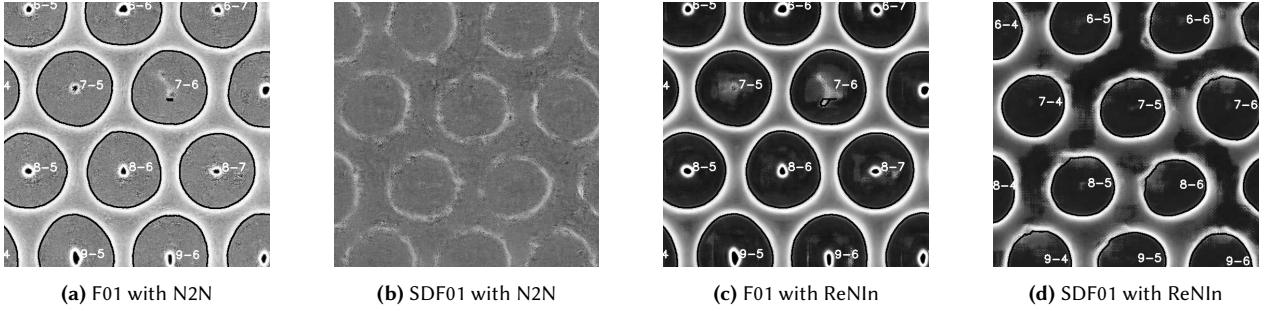


Fig. 8. Qualitative results on the circle detection. The circle detection is performed over the image [Figure 6](#) and [Figure 7](#). Their results are indicated by the black circle line above the white edge (For better understanding, please check [Figure 6](#) and [Figure 7](#)). Due to the circle line, the color of the images may appear shifted, but this is purely a visualization difference and does not affect the material properties. The numbers within the images indicate the circle numbering, ranging from 1-1 to 9-12. It is used to identify each NAND cell for further inspection.

Table 4. Ablation results of ReNIn. Table shows each component of ReNIn: Relaxed Noise2Noise (RN2N) and Input dropout (ID). We evaluate the models in two different architectures: FBI-Net and UNet. FBI-Net and UNet are representative model architectures for BSN and non-BSN, respectively.

Model	+RN2N	+ID	PSNR	SSIM
FBI-Net	X	X	17.65	0.1848
	O	X	18.81	0.2400
	X	O	19.78	0.2545
	O	O	20.85	0.2712
UNet	X	X	18.09	0.2139
	O	X	18.76	0.2328
	X	O	19.80	0.2466
	O	O	20.35	0.2636

of noise correlation in our SEM images, which violates the core assumption of those methods. This will be further discussed in the [Noise correlation in SEM images](#) section.

Ablation studies on modeling choices

[Table 4](#) presents the effectiveness of the proposed components, Relaxed Noise2Noise (RN2N) and Input Dropout (ID), across different model architectures: FBI-Net (BSN) and UNet (non-BSN). Following the [Lehtinen et al. \(2018\)](#), we use their implementation of UNet architecture. Regarding batch size in training, we use batch size of 1 for FBI-Net and 4 for UNet following their implementations. The base model (Noise2Noise, F01-F01) is shown in rows 1 and 5. Both RN2N (rows 2 and 6) and ID (rows 3 and 7) individually improve denoising performance, showing their effectiveness and compatibility with different architectures. Lastly, when both components are combined (rows 4 and 8, our final method, ReNIn), there is a further improvement in performance. Furthermore, we observe there is no significant performance dif-

Table 5. Target images with different noise levels. Here, all variants apply the input dropout (ID), with only the noise level of target images differing.

Model variants	PSNR	SSIM	FR
F01-F01 (N2N + ID)	19.78	0.2545	7.56%
F01-F02	19.96	0.2570	5.25%
F01-F08	20.85	0.2712	1.08%
F01-F16	20.87	0.2806	2.62%
F01-F64 (Sup + ID)	20.80	0.2939	1.54%

ference between the BSN and non-BSN architecture¹.

More analysis on ReNIn

In this subsection, we show a more detailed analysis on ReNIn.

[Target images with different noise levels] [Table 5](#) compares variants of ReNIn which utilize the different noise levels of target images, such as F02 and F16 (F08 images are used in our main experiment). All variants incorporate the input dropout (ID), with the only change being the target images. For simplicity, we use the notation F#1-F#2 to denote the training scheme, where F#1 represents input images and F#2 represents target images. Specifically, F01-F01 refers to “Noise2Noise with input dropout”, while F01-F64 denotes the “supervised learning method with input dropout”. We observe incremental performance improvements, particularly in SSIM, as target images become less noisy. It highlights the practical usability of ReNIn, which provides broader training options regarding both data collection cost and denoising performance.

[Impact of the varying dropout ratio] We assess the impact of different dropout ratios—the percentage of masked pixels in the input image—on denoising performance. Here,

¹Note that the supervised learning method also has no improvement with non-BSN architecture.

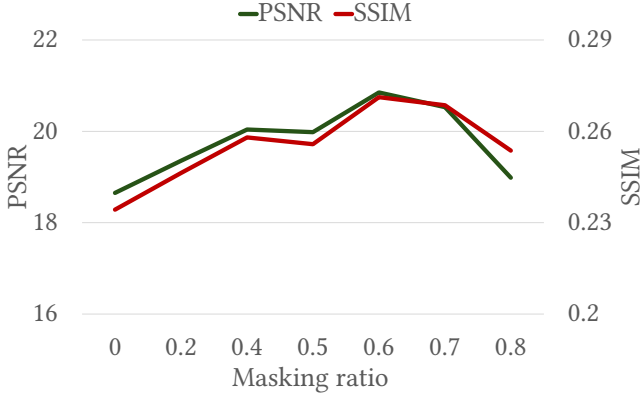


Fig. 9. Impact of the varying dropout ratio. F01-F08 model is used as a base model (*i.e.*, all models use the F08 image as the target). The dropout ratio indicates the proportion of the masked pixel among all the pixels. Note that the zero masking ratio indicates the F01-F08 model without input dropout.

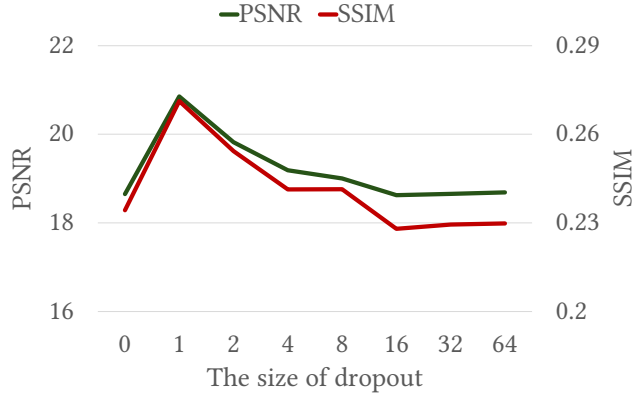


Fig. 10. Impact of the varying dropout patch size. This figure follows the same details as Figure 9, using the F01-F08 model as the base. A dropout patch size of 0 indicates training without input dropout, while a size of 1 corresponds to the pixel-wise dropout strategy.

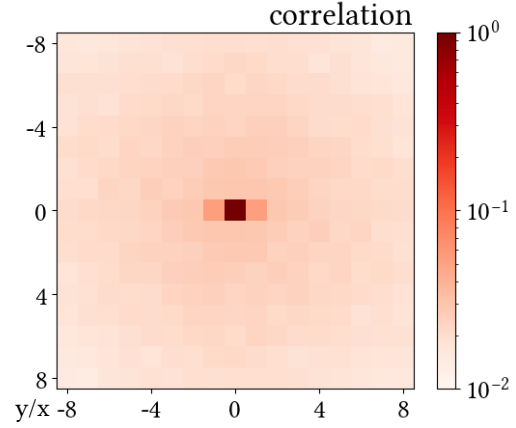


Fig. 11. The spatial correlation map between pixels based on relative positions. The x-axis denotes horizontal positions and the y-axis denotes vertical ones. Each coordinate represents the correlation between the pairs of pixels that differ by x and y . For example, the (1,2) of (x,y) of coordinate indicates the correlation between pairs of pixels that are 1 pixel apart horizontally and 2 pixels apart vertically. Note that the (0,0) coordinate indicates the correlation between the pixel and itself.

the F01-F08 model is used as a base model (*i.e.*, all models use the F08 image as the target). As shown in Figure 9, we test various ratios and identify the probability of 0.6 as the most effective, consistently yielding the highest performance in both PSNR and SSIM metrics. This ratio was therefore selected as the optimal setting for our final model.

[Impact of varying dropout patch size] We explore the alternative strategies that drop patches, rather than individual pixels, within the image. To analyze this, we conduct experiments with various patch dropout sizes, ranging from 0 to 64. Here, a patch size of 2 refers to a 2×2 area of pixels being dropped. A dropout size of 0 indicates no dropout, while a size of 1 corresponds to the pixel-wise dropout used in the main experiments. As shown in Figure 10, we observe that pixel-wise dropout where the dropout size is one pixel consistently outperforms other dropout strategies.

Discussion

Noise correlation in SEM images

We observe that self-supervised learning methods (Noise2Void and FBI-denoiser) exhibit significantly inferior denoising performance in Table 2. We hypothesize that this is primarily due to the noise in our SEM images not being pixel-wise independent. To verify this, we conduct noise correlation analysis following Lee et al. (2022). We calculate the Pearson correlation coefficient between the pairs of noise pixels with a difference of x and y (horizontal and vertical direction, respectively). Here, the noise was

approximated by calculating the difference between F01 and F64 images. As shown in Figure 11, the correlation analysis reveals a relatively high correlation between adjacent neighboring pixels on the left and right, suggesting a spatial correlation. This highlights discrepancies between the actual characteristics of our SEM image noise and the noise assumptions in self-supervised learning methods, which assume noise is conditionally pixel-wise independent given the clean images. We believe these discrepancies lead to the inferior denoising performance observed in self-supervised learning methods.

Conclusions

Our paper presents ReNIn which effectively addresses key challenges in SEM image denoising by offering an efficient balance between denoising performance and data collection cost through the Relaxed Noise2Noise (RN2N) framework. Additionally, the integration of input dropout strategy enhances generalization to unseen image structures without compromising performance on familiar ones, leading to a reduced failure rate in circle detection—a critical step in product analysis for semiconductor manufacturing. Overall, ReNIn offers an efficient and robust solution for SEM image denoising. For more details and access to our implementation, please visit the [Project page](#).

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Conflict of Interest

Competing interests: Author YongWoo Lee is employed at company Samsung.

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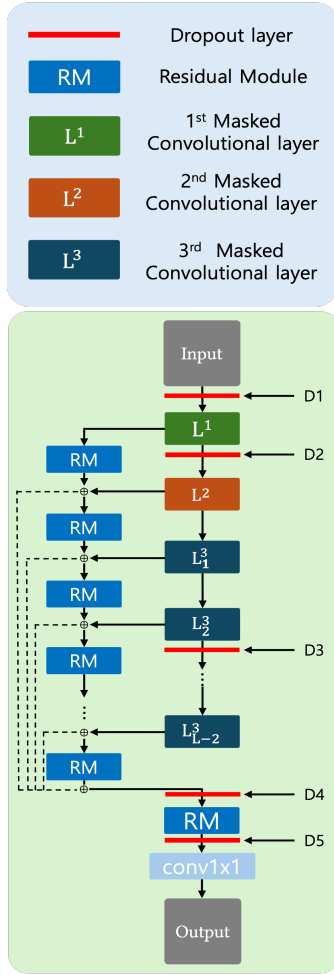


Fig. 12. The architecture of FBI-Net and dropout position from D1 to D6. The details of the masked convolutional layer and residual module can be found in [Byun et al. \(2021\)](#)

Appendix

Dropout experiment details

To evaluate the effect of the dropout layer, we follow the same setting as our main experiment. The only difference is in the FBI-Net network architecture which adopts an additional dropout layer within the model. Similar to [Kong et al. \(2022\)](#), we choose five different locations to add the dropout layer. These are marked from D1 to D5, which are represented in [Figure 12](#). We implement a dropout layer with PyTorch framework ([Paszke et al., 2019](#)). The details of FBI-Net architecture such as the masked convolutional layer and residual module are described in [Byun et al. \(2021\)](#).

PSNR matching

To address the problem of pixel mismatch between images depicting the same scene, we propose an intuitive and

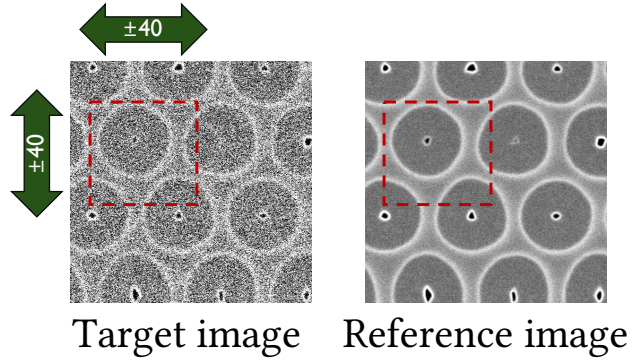


Fig. 13. A simplified overview of PSNR matching. For clarity, we demonstrate using image patches rather than the entire image.

straightforward algorithm known as PSNR matching. This algorithm identifies pixel shift information by locating the best-matching patches using the PSNR metric. It is quite effective when only pixel shift occurs between two images of the identical scene. The algorithm consists of three steps:

1. Selecting appropriate patches with identical positions in the images.
2. Fixing the reference patch (typically the cleaner patch is chosen) and shifting the other patch horizontally and vertically.
3. Compute the PSNR for each shifted patch to identify the position with the highest PSNR, which indicates the best matching position.

In detail, we choose the reference patch from F64 images and randomly select the position to crop the patch, as the image displays similar patterns repeatedly. We opt for a crop size of 256, and the search range to shift vertically and horizontally is set to 40. The [Figure 13](#) illustrates a simple overview of this algorithm.

With this algorithm, we align the images to match appropriate pixel positions within the scene. However, there is a loss of information near the borders, where the search range multiplied by 2 represents the maximum number of lost pixels.